

Sahithi Palumari

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PROFESSIONAL SUMMARY

Senior Data Analyst with 5+ years of experience delivering actionable insights and data-driven solutions across Healthcare Financial Services, and Education sectors. Expertise in SQL, Python, R, Tableau, Power BI, and cloud analytics with proven track record of building scalable data pipelines, predictive models, and executive dashboards that drive measurable business outcomes. Strong proficiency in ETL automation, statistical analysis, and translating complex data into compelling narratives for stakeholders. Demonstrated success in optimizing operations, reducing costs up to 50%, and improving decision-making speed by 40% through advanced analytics and visualization.

TECHNICAL SKILLS

Database & Querying: SQL (Oracle, MySQL, PostgreSQL), NoSQL (MongoDB, Cassandra, DynamoDB)

Data Visualization: Tableau, Power BI, QlikView

Programming: Python (Pandas, NumPy, Matplotlib, Seaborn), R

Statistical Analysis: Hypothesis Testing, Regression Analysis, Time Series Forecasting

ETL & Data Management: Informatica, SSIS, Hadoop

Cloud Platforms: AWS (S3, Glue, Redshift), Azure (Data Factory, Blob Storage, Data Lake)

Excel: Advanced Excel (Pivot Tables, Power Query, VBA)

Version Control: Git, GitHub, Bitbucket

Testing & Validation: Data Quality Checks, Data Validation, Unit Testing (pytest)

Security & Governance: Data Governance, Role-Based Access Control

Skills: Data Storytelling, Problem Solving, Stakeholder Communication

WORK EXPERIENCE

Cigna Healthcare | Sr Data Analyst – Healthcare Analytics | Chicago, USA

Mar 2024 – Present

Responsibilities:

- Engineered and optimized complex SQL queries on Oracle databases processing multi-terabyte healthcare claims data (500M+ records annually), accelerating data extraction and reporting performance by 60% through advanced indexing strategies, query tuning, and database optimization techniques.
- Built interactive Tableau dashboards visualizing critical KPIs including claim turnaround time, approval rates, denial patterns, provider performance, and member demographics across 15+ business units, enabling leadership to identify operational bottlenecks and implement improvements reducing claim processing time by 28%.
- Automated ETL pipelines using Azure Data Factory, Python, and SSIS reducing manual data preparation time by 80%, integrating data from 50+ disparate sources including claims systems, EHR platforms, pharmacy databases, and third-party vendors ensuring data quality and consistency.
- Performed exploratory data analysis (EDA) using Python (Pandas, NumPy, Matplotlib) on patient admission patterns, emergency room utilization, and readmission trends, uncovering insights that enabled hospitals to optimize staffing schedules, reduce wait times by 25%, and improve resource allocation saving \$12M annually.
- Partnered with compliance teams to implement HIPAA-compliant data anonymization and de-identification techniques using Python and Informatica, protecting PHI (Protected Health Information) while preserving analytical value for 18M+ patient records and ensuring regulatory compliance with federal guidelines.
- Designed predictive models using Python (Scikit-learn) and R for hospital readmission risk, high-cost patient identification, and chronic disease management, achieving 85% accuracy and enabling proactive care coordination that reduced readmissions by 22% and medical costs by \$8M.
- Deployed secure Azure analytics environments using Terraform and Azure DevOps supporting scalable data processing, implementing role-based access control (RBAC), encryption, and data governance policies ensuring compliance with HIPAA and organizational security standards.
- Conducted root cause analysis of ETL pipeline failures and data quality issues, implementing process improvements including automated testing (pytest), data validation frameworks, and monitoring dashboards that elevated overall data accuracy from 92% to 98.5%.

Environment: SQL, Python, R, Oracle, Power BI, Tableau, Azure Data Factory, Azure Synapse, AWS S3, SSIS, Informatica, Git, Terraform, Pandas, NumPy, Scikit-learn, Excel, SPSS, HIPAA Compliance, JIRA, Confluence, ETL, Data Quality, Predictive Modeling.

American Express | Financial Analyst | Hyderabad, India

Mar 2021 – Dec 2022

Responsibilities:

- Delivered financial analytics supporting credit card operations, fraud detection, and risk management for American Express serving 130+ million card members globally, building SQL-based reports and Power BI dashboards that provided real-time visibility into credit behavior, transaction patterns, and risk trends.

- Designed and optimized complex SQL queries on MySQL databases analyzing 2 billion+ annual transactions, supporting fraud detection workflows and reducing investigation turnaround time by 30% through automated data extraction, risk scoring, and anomaly identification.
- Developed Power BI dashboards visualizing fraud patterns, transaction velocity, geographic anomalies, and merchant risk profiles across 200+ countries, empowering fraud investigation teams to proactively mitigate risks that prevented \$45M in fraudulent losses annually. Automated data extraction, cleansing, and transformation pipelines using Python (Pandas) and AWS Glue, eliminating manual bottlenecks reducing processing time by 75%, and ensuring timely delivery of daily, weekly, and monthly financial reports to senior management and regulatory bodies.
- Migrated legacy ETL workflows from on-premise infrastructure to AWS cloud platform using AWS Glue, S3, and Redshift, enabling real-time analytics capabilities, improving scalability to handle 10X data volume growth, and reducing infrastructure costs by 35%.
- Built statistical models in R and Python predicting credit default probability using logistic regression, random forests, and gradient boosting, analyzing 50+ risk factors and achieving 83% prediction accuracy, enhancing risk scoring models and reducing potential credit losses by \$25M.
- Implemented SQL-Python anomaly detection framework using statistical methods and machine learning algorithms to identify suspicious transactions, unusual spending patterns, and account takeover attempts, improving fraud monitoring precision by 40% and reducing false positives by 35%.
- Optimized database performance through query tuning, indexing strategies, and partition management on MySQL reducing report execution time by 55% during peak financial cycles (month-end, quarter-end) and improving user satisfaction scores from 72 to 91.
- Performed customer segmentation analysis using clustering algorithms (K-means) in Python, identifying 12 distinct customer personas based on spending behavior, credit utilization, and demographic factors, enabling personalized marketing strategies improving campaign response rates by 35%.
- Automated monthly financial reconciliation reports using advanced Excel (VBA, Power Query) and Python, reducing manual reporting time by 50%, eliminating human errors, and strengthening audit readiness for internal compliance reviews and regulatory examinations. Produced comprehensive data lineage documentation and data dictionaries supporting compliance audits and regulatory examinations, ensuring transparency in reporting processes and maintaining 100% audit compliance rate across quarterly reviews.

Environment: SQL, Python, R, MySQL, Power BI, AWS (S3, Glue, Redshift, Athena), Informatica, Pandas, NumPy, Scikit-learn, Excel, VBA, Git, SPSS, Fraud Detection, Risk Analytics, Statistical Modeling, ETL.

Virtusa | Business Intelligence-Data Analyst | Hyderabad, India

Oct 2019 – Mar 2021

Responsibilities:

- Managed end-to-end data analytics for higher education institution serving 25,000+ students and 1,500+ faculty, collecting and integrating data from Oracle SQL, Banner ERP, Student Information Systems (SIS), and third-party vendor platforms to build comprehensive datasets supporting institutional reporting and strategic planning.
- Developed and maintained institutional data warehouse using Oracle SQL creating 150+ custom data elements, dimensional tables, and fact tables supporting enrollment analytics, student outcomes, faculty performance, and financial aid reporting for administrative decision-making. Merged and integrated multiple internal and external datasets using SQL joins, data mapping, and ETL workflows ensuring data accuracy, consistency, and scalability across student demographics, course registrations, grades, financial aid, and alumni engagement databases.
- Performed comprehensive quality assurance checks and data validation using Python and SQL, implementing automated testing frameworks that verified data completeness, accuracy, referential integrity, and consistency reducing data errors by 65%.
- Built Tableau dashboards and reports visualizing student enrollment trends, retention metrics by demographics, course completion rates, financial aid distribution, and academic performance patterns enabling academic leadership to make data-driven decisions on resource allocation and program improvements.
- Translated complex statistical findings into clear, actionable recommendations delivering presentations and executive reports in non-technical language to provosts, deans, department chairs, and administrative staff supporting strategic initiatives and accreditation requirements. Partnered with IT teams and Project Management Office to design and implement enterprise data warehouse architecture aligned with institutional requirements, integrating 20+ disparate systems and establishing data governance policies ensuring data quality and security.
- Created master course schedules by consolidating campus-level and online program offerings using SQL queries and Python scripts, streamlining academic planning workflows and reducing scheduling conflicts by 40% while optimizing classroom utilization and faculty assignments.

Environment: SQL, Python, Oracle SQL, Banner ERP, SPSS, Tableau, Excel, Git, ETL, Data Warehouse, Statistical Analysis, Higher Education Analytics.

Education

Eastern Illinois University

Master of Science, Computer Technology

Dec 2024